

Element E: Application of STEM Principles and Practices

E1. Application of STEM Principles to Design Requirements

- **Design Requirements Overview:** Briefly list your high-priority design requirements.
 1. The design must be able to store up to 200 rolls of 3d printer filament.
 2. Must be able to support up to 200 rolls weighing 1 kg each without breaking or sinking.
 3. Rolls must be organized by their specific color of filament, size of rolls, thickness of filament, and material of filament.
 4. The design needs to store the rolls out of the way and without interfering with any other important places/things in the room.
- **STEM Applications:** Describe how specific STEM principles and practices apply to these requirements.
 - **Example Application:** Provide a clear example where a STEM principle enhances a specific requirement.

Going off of the STEM principle of using falsifiable and testable conditions, and being able to plan and carry out investigations/experiments, we made each of these design requirements so that they could be adequately tested in a prototype to analyze how effective that prototype is, and tweak it if necessary. For design requirement 1, being able to store 200 rolls of 3D printer filament is easily testable. We could either use 200 of Mr. L's filaments to directly test this, or test this by repeatedly stacking up the few filaments we had from Mr. Gregory's room (we had enough filaments to totally fill at least one dowel) on every dowel, and counting up how many the design could stack in total. The same principles of investigation and testing apply to requirement 2, where we could again use 200 of Mr. L's filaments to test whether the prototype breaks.

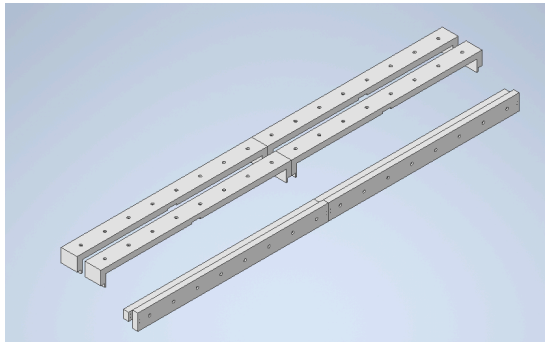
- **Calculations and Theories:** Include relevant calculations or theories that support the application.

Popper's falsification theory: <https://www.simplypsychology.org/karl-popper.html>

For design requirement 1, the prototypes we design would all have 48 dowels of the same length, so if one dowel were able to support at least 5 dowels ($200/48 = 4.167$), then we can easily test the requirement using just one dowel. For design requirement 2, the prototype must be able to support $200 \text{ dowels} \times 1 \text{ kg/dowel} = 200 \text{ kg}$, so one dowel section has to be able to support $200 \text{ kg}/48 = 4.167 \text{ kg per dowel section}$.

E2. Substantiation of Proposed Solution with STEM

- **Solution Overview:** Outline your proposed solution.



All the spools will be stored on the top and side shelves on vertical and horizontal pegs organized by size, color, or whatever criterion Mr. L wants it to be organized by. The top shelf will be built from plywood and laid across the existing scaffold, which is 144 in x 21 in, so the actual plywood will need to be about an inch smaller on each side. Vertical pegs will have a diameter of 1 inch and will be spaced 9 inches apart, making a grid of 16 x 2 vertical pegs on the shelf, each peg is 16 inches tall. The vertical pegs will serve as long term storage that can store the bulk of the spools. The horizontal pegs will serve as short term storage to allow students to have easy access to the spools to swap out with the existing spools. There will be one horizontal shelf compared to two vertical shelves. The pegs will be 1.5 inches in diameter.

- **STEM Justification:** Explain how STEM principles substantiate your solution.
 - **Detailed Substantiation:** Provide a rationale for each STEM principle used, linking to the solution's effectiveness.

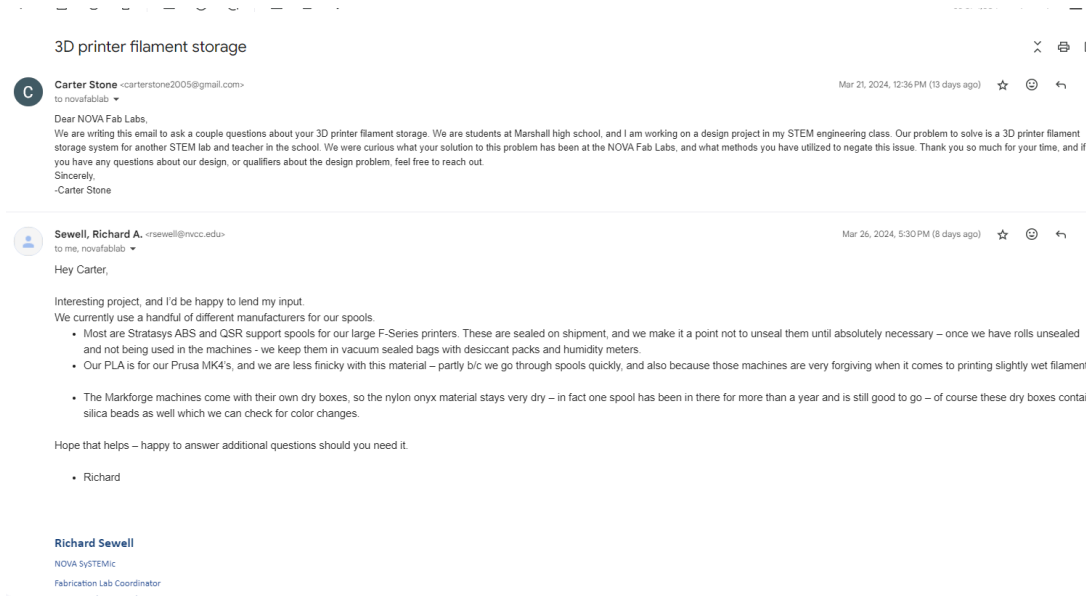
Mathematical calculations/principles were used to determine the optimal size of the shelf and the spacing of the pegs. The 9-inch spacing and the 16-inch height of the pegs were calculated to maximize the number of spools stored while still ensuring ease of access. We similarly used mathematical calculations to ensure that the (3 separate) plywood beams would be able to support $200/3 = 67$ dowels. According to engineering principles, we designed the peg system to be modular, allowing for easy reconfiguration as needed.

E3. Expert Review of STEM Applications

- **Expert Consultation:** Summarize the outcome of expert consultations regarding your STEM applications.
 - **Feedback Summary:** Highlight key feedback that influenced further development or confirmation of your design.

E4. Documentation of Expert Credentials and Review

- **Expert Credentials:** We got the review from Richard Sewall, the Fabrication Lab Coordinator at Nova Fab Labs, an engineering lab at NOVA community college in northern Virginia.
- **Review Documentation:** Provide evidence of expert review, such as excerpts or summaries.



E5. Impact of Expert Review on Design

- **Verification and Corrective Responses:** Describe how expert reviews confirmed or led to modifications in your design.
 - **Changes Made:** List specific changes made based on expert feedback and the rationale behind them.

From the expert review we got a good idea of what solution to filament storage they use at NOVA Fab Labs, and we adapted our design accordingly. We kept the actual storage system very close to the 3D printers for the convenience of everyone in the classroom in the design. This decision was based on how they use boxes close to the printers at NOVA Fab Labs, and the convenience that it brings. We switched to dowels/pipes that can be screwed into the shelf using 3D printed connections in the design, also for ease of access which was emphasized in the feedback of the expert reviewer and what they do in their labs. We switched from wooden dowels to cheaper and lighter PVC, which will work better for our design in this context, and which is validated by the use of the cheapest possible materials for storage in the labs.

Conclusion Reflect briefly on integrating STEM principles into your project, focusing on the impact on your design process and the challenges addressed.

We used STEM principles such as ability to investigate and falsify and using mathematical calculation and engineering methods to inform the design requirements and the design itself. These had a large impact on our design requirements in terms of testability, falsifiability, and measurability. Reflecting on the design requirements, it is clear that good design requirements should conform to this scientific principle in order to make a good final design. The principles also had a large impact on the design in terms of the materials used, dimensions used, and the modularity of the design.